MAT 3007 – Optimization

Final Review

Instructor: Junchi Yang

CUHK-Shenzhen

Review

The second half of the semester mainly covers nonlinear optimization and integer optimization:

- Optimality conditions for nonlinear optimization.
- Convexity.
- Algorithms.
- Integer optimization

Review

- Optimality conditions for nonlinear optimization.
  - Unconstrained problems: FONC, SONC, SOSC.
  - Constrained problems: KKT conditions.
- Convexity.
- Algorithms.
- Integer optimization

First-Order Necessary Conditions

If $\boldsymbol{x}^*$ is a local minimizer of the unconstrained problem $\min_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x})$, then we must have $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$.

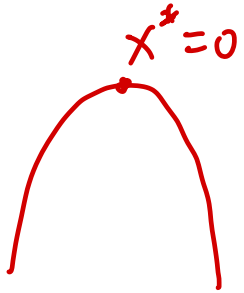
First-Order Necessary Conditions

If $\boldsymbol{x}^*$ is a local minimizer of the unconstrained problem $\min_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x})$, then we must have $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$.

Remark: This is not a sufficient condition. Consider

$$f(x) = -x^2$$

at $x = 0$.



$f'(0) = 0$

Local max and saddle points also need to satisfy $\nabla f(x^*) = 0$.

Theorem: Second-Order Necessary Conditions

If $\boldsymbol{x}^*$ is a local minimizer of f, then it holds that:

1. $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$;
2. $\nabla^2 f(\boldsymbol{x}^*)$ is positive semi-definite.

Theorem: Second-Order Necessary Conditions

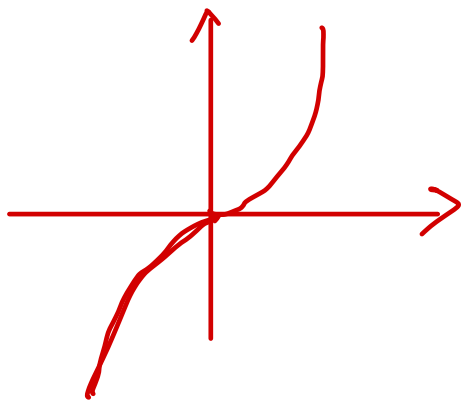
If $\boldsymbol{x}^*$ is a local minimizer of f, then it holds that:

1. $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$;
2. $\nabla^2 f(\boldsymbol{x}^*)$ is positive semi-definite.

Remark: This is not a sufficient condition. Consider

$$f(x) = x^3$$

at $x = 0$.



$f'(0) = 0$, $f''(0) = 0$

but $x=0$ is not local min.

Theorem: Second-Order Sufficient Conditions

Let f be twice continuously differentiable. If $\boldsymbol{x}^*$ satisfies:

1. $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$;
2. $\nabla^2 f(\boldsymbol{x}^*)$ is positive definite.

then $\boldsymbol{x}^*$ is a strict local ~~minimum~~ minimizer of f.

Theorem: Second-Order Sufficient Conditions

Let f be twice continuously differentiable. If $\boldsymbol{x}^*$ satisfies:

1. $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$;
2. $\nabla^2 f(\boldsymbol{x}^*)$ is positive definite.

then $\boldsymbol{x}^*$ is a strict local minimum of f.

Remark: This is not a necessary condition. Consider

$$f(x) = x^4$$

at $x = 0$. $f'(x) = 4x^3$ $f''(x) = 12x^2$

$f'(0) = 0$ $f''(0) = 0$ $\longrightarrow$ does not satisfy SOSC.

But $x=0$ is a strict global min because $x^4 > 0$ for $x \neq 0$.

Indefiniteness and Saddle Points

Definition: Stationary Points and Saddle Points

- A point $\boldsymbol{x}$ satisfying $\nabla f(\boldsymbol{x}) = \mathbf{0}$ is called critical point or stationary point.
- A stationary point is called saddle point if it is neither a local minimizer nor a local maximizer.

If $\nabla f(x) = 0$, then x can be local min, local max or saddle point.

Indefiniteness and Saddle Points

Definition: Stationary Points and Saddle Points

- A point $\boldsymbol{x}$ satisfying $\nabla f(\boldsymbol{x}) = \mathbf{0}$ is called critical point or stationary point.
- A stationary point is called saddle point if it is neither a local minimizer nor a local maximizer.

Corollary: A Sufficient Condition for Saddle Points

Suppose that $\boldsymbol{x}^*$ is a stationary point ($\nabla f(\boldsymbol{x}^*) = \mathbf{0}$) and that the Hessian $\nabla^2 f(\boldsymbol{x}^*)$ is indefinite, then $\boldsymbol{x}^*$ is a saddle point.

If a stationary point x satisfies

① SOSC for local min $\Rightarrow$ strict local min
($\nabla^2 f(x)$ is PD)

② SOSC for local max $\Rightarrow$ strict local max
($\nabla^2 f(x)$ is ND)

③ $\nabla^2 f(x)$ is indefinite $\Rightarrow$ saddle point

Remark: All of these are sufficient condition. For the case such as $\nabla^2 f(x)$ is PSD (but not PD), it may not be easy to classify.

Definite Matrices

Definiteness: Let $\boldsymbol{A}$ be a real $n \times n$ matrix.

- $\boldsymbol{A}$ is called positive (negative) semidefinite if $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} \geq 0$ (≤ 0) for all $\boldsymbol{x} \in \mathbb{R}^n$.
- The matrix $\boldsymbol{A}$ is positive (negative) definite if $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} > 0$ (< 0) for all $\boldsymbol{x} \in \mathbb{R}^n \backslash \{\boldsymbol{0}\}$.
- $\boldsymbol{A}$ is said to be indefinite if $\boldsymbol{A}$ is neither positive semidefinite nor negative semidefinite.

Definite Matrices

Definiteness: Let $\boldsymbol{A}$ be a real $n \times n$ matrix.

- $\boldsymbol{A}$ is called positive (negative) semidefinite if $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} \geq 0$ (≤ 0) for all $\boldsymbol{x} \in \mathbb{R}^n$.
- The matrix $\boldsymbol{A}$ is positive (negative) definite if $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} > 0$ (< 0) for all $\boldsymbol{x} \in \mathbb{R}^n \backslash \{\boldsymbol{0}\}$.
- $\boldsymbol{A}$ is said to be indefinite if $\boldsymbol{A}$ is neither positive semidefinite nor negative semidefinite.

Theorem: Eigenvalues and Definiteness

Let $\boldsymbol{A}$ be a real symmetric $n \times n$ matrix with eigenvalues $\lambda_i \in \mathbb{R}$, $i = 1, \ldots, n$. It follows:

- $\boldsymbol{A}$ is positive (negative) semidefinite iff $\lambda_i \geq 0$ (≤ 0) for all i.
- $\boldsymbol{A}$ is positive (negative) definite iff $\lambda_i > 0$ (< 0) for all i.
- $\boldsymbol{A}$ is indefinite iff there are $i, j \in \{1, \ldots, n\}$ with $\lambda_i > 0$ and $\lambda_j < 0$.

Review

- Optimality conditions for nonlinear optimization.
  - Unconstrained problems: FONC, SONC, SOSC.
  - Constrained problems: KKT conditions.
- Convexity.
- Algorithms.
- Integer optimization

Abstract FONC for Constrained Problems

Definition: Feasible Direction

Given $\boldsymbol{x} \in \Omega$, we call $\boldsymbol{d}$ a feasible direction at $\boldsymbol{x}$ if there exists $\bar{t} > 0$ such that $\boldsymbol{x} + t\boldsymbol{d} \in \Omega$ for all $0 \leq t \leq \bar{t}$.

Abstract FONC for Constrained Problems

Definition: Feasible Direction

Given $\boldsymbol{x} \in \Omega$, we call $\boldsymbol{d}$ a feasible direction at $\boldsymbol{x}$ if there exists $\bar{t} > 0$ such that $\boldsymbol{x} + t\boldsymbol{d} \in \Omega$ for all $0 \leq t \leq \bar{t}$.

Definition: Descent Direction

Let f be continuously differentiable. Then $\boldsymbol{d}$ is called a descent direction at $\boldsymbol{x}$ if and only if $\nabla f(\boldsymbol{x})^\top \boldsymbol{d} < 0$.

Abstract FONC for Constrained Problems

Definition: Feasible Direction

Given $\boldsymbol{x} \in \Omega$, we call $\boldsymbol{d}$ a feasible direction at $\boldsymbol{x}$ if there exists $\bar{t} > 0$ such that $\boldsymbol{x} + t\boldsymbol{d} \in \Omega$ for all $0 \leq t \leq \bar{t}$.

Definition: Descent Direction

Let f be continuously differentiable. Then $\boldsymbol{d}$ is called a descent direction at $\boldsymbol{x}$ if and only if $\nabla f(\boldsymbol{x})^\top \boldsymbol{d} < 0$.

Theorem: FONC for Constrained Problems

Let $\boldsymbol{x}^*$ be a local minimum of $\min_{\boldsymbol{x} \in \Omega} f(\boldsymbol{x})$. Then for any feasible direction $\boldsymbol{d}$ at $\boldsymbol{x}^*$, we must have $\nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d} \geq 0$.

Abstract FONC for Constrained Problems

Definition: Feasible Direction

Given $\boldsymbol{x} \in \Omega$, we call $\boldsymbol{d}$ a feasible direction at $\boldsymbol{x}$ if there exists $\bar{t} > 0$ such that $\boldsymbol{x} + t\boldsymbol{d} \in \Omega$ for all $0 \leq t \leq \bar{t}$.

Definition: Descent Direction

Let f be continuously differentiable. Then $\boldsymbol{d}$ is called a descent direction at $\boldsymbol{x}$ if and only if $\nabla f(\boldsymbol{x})^\top \boldsymbol{d} < 0$.

Theorem: FONC for Constrained Problems

Let $\boldsymbol{x}^*$ be a local minimum of $\min_{\boldsymbol{x} \in \Omega} f(\boldsymbol{x})$. Then for any feasible direction $\boldsymbol{d}$ at $\boldsymbol{x}^*$, we must have $\nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d} \geq 0$.

Remark:

- This abstract FONC means that there are no feasible descent directions.
- In the unconstrained case, all directions are feasible. Thus, we must have $\nabla f(\boldsymbol{x}^*) = \boldsymbol{0}$.

$\nabla f(x^*)^T d \geq 0, \forall d \Rightarrow \nabla f(x^*) = 0$

Setup :

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \leq 0, \quad \forall\ i = 1, \ldots, m, \\ & h_j(\boldsymbol{x}) = 0, \quad \forall\ j = 1, \ldots, p. \end{aligned}$$

Setup :

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \le 0, \quad \forall\ i = 1, \dots, m, \\ & h_j(\boldsymbol{x}) = 0, \quad \forall\ j = 1, \dots, p. \end{aligned}$$

- The feasible set is $\Omega = \{\boldsymbol{x} \in \mathbb{R}^n : g(\boldsymbol{x}) \le \mathbf{0},\ h(\boldsymbol{x}) = \mathbf{0}\}$.
- For $\boldsymbol{x} \in \Omega$, the set $\mathcal{A}(\boldsymbol{x}) := \{i : g_i(\boldsymbol{x}) = 0\}$ denotes the set of active constraints at $\boldsymbol{x}$. The set of inactive constraints is $\mathcal{I}(\boldsymbol{x}) := \{i : g_i(\boldsymbol{x}) < 0\}$.

Setup :

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \le 0, \quad \forall\ i = 1, \dots, m, \\ & h_j(\boldsymbol{x}) = 0, \quad \forall\ j = 1, \dots, p. \end{aligned}$$

- The feasible set is $\Omega = \{\boldsymbol{x} \in \mathbb{R}^n : g(\boldsymbol{x}) \le \boldsymbol{0},\ h(\boldsymbol{x}) = \boldsymbol{0}\}$.
- For $\boldsymbol{x} \in \Omega$, the set $\mathcal{A}(\boldsymbol{x}) := \{i : g_i(\boldsymbol{x}) = 0\}$ denotes the set of active constraints at $\boldsymbol{x}$. The set of inactive constraints is $\mathcal{I}(\boldsymbol{x}) := \{i : g_i(\boldsymbol{x}) < 0\}$.

Lagrangian: λ, μ : Lagrangian multipliers

$$L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) := f(\boldsymbol{x}) + \sum_{i=1}^{m} \lambda_i g_i(\boldsymbol{x}) + \sum_{j=1}^{p} \mu_j h_j(\boldsymbol{x})$$

Necessary Condition – KKT Conditions

If $\boldsymbol{x}$ is a local minimizer and a Constraint Qualification ($\star$) holds, then there exist $\boldsymbol{\lambda}$ and $\boldsymbol{\mu}$ such that:

1. Main Condition

$$\nabla_x L(x,\lambda,\mu) = \nabla f(\boldsymbol{x}) + \sum_{i=1}^{m} \lambda_i \nabla g_i(\boldsymbol{x}) + \sum_{j=1}^{p} \mu_j \nabla h_j(\boldsymbol{x}) = \mathbf{0}.$$

2. Dual Feasibility

$$\lambda_i \geq 0 \quad i = 1, \ldots, m.$$

3. Complementarity

$$\lambda_i \cdot g_i(\boldsymbol{x}) = 0 \quad \forall\ i = 1, \ldots, m.$$

We often add primal feasibility as part of the KKT conditions:

4. Primal Feasibility

$$g_i(\boldsymbol{x}) \leq 0, \quad h_j(\boldsymbol{x}) = 0 \quad \forall\ i, \quad \forall\ j.$$

Linear Independence Constraint Qualification (LICQ): We require the collection of gradients

$$\{\nabla g_i(\boldsymbol{x}) : i \in \mathcal{A}(\boldsymbol{x})\} \cup \{\nabla h_j(\boldsymbol{x}) : j = 1, \ldots, p\} \qquad (\star)$$

to be linearly independent.

LICQ and KKT Points

Linear Independence Constraint Qualification (LICQ): We require the collection of gradients

$$\{\nabla g_i(\boldsymbol{x}) : i \in \mathcal{A}(\boldsymbol{x})\} \cup \{\nabla h_j(\boldsymbol{x}) : j = 1, \ldots, p\} \qquad (\star)$$

to be linearly independent.

Remark:

- A feasible point $\boldsymbol{x}$ satisfying the LICQ is called regular.
- A (feasible) point satisfying the KKT conditions is called a KKT point.

Sometimes, we can use KKT conditions to find global minimizer (for some very simple case):

① find out all KKT points. (trick: split into subcases whether $\lambda_i < 0$ or $= 0$ / or whether $g_i < 0$ or $= 0$)

② compare function values of those KKT points

③ check whether there are points that do not satisfied LICQ.

KKT Conditions with Linear Constraints

Consider an inequality constrained problem:

$$\text{minimize}_{\boldsymbol{x}} \; f(\boldsymbol{x}) \quad \text{s.t.} \quad \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}. \tag{1}$$

Theorem: KKT for Linearly Constrained Problems

If $\boldsymbol{x}^*$ is a local minimizer of (1), then there exists some $\boldsymbol{y} \geq \boldsymbol{0}$ with

$$\nabla f(\boldsymbol{x}^*) + \boldsymbol{A}^\top \boldsymbol{y} = \boldsymbol{0}$$
$$y_i \cdot (\boldsymbol{a}_i^\top \boldsymbol{x}^* - b_i) = 0 \quad \forall \; i,$$

where $\boldsymbol{a}_i^\top$ is the ith row of $\boldsymbol{A}$.

KKT Conditions with Linear Constraints

Consider an inequality constrained problem:

$$\text{minimize}_{\boldsymbol{x}} \; f(\boldsymbol{x}) \quad \text{s.t.} \quad \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}. \tag{1}$$

Theorem: KKT for Linearly Constrained Problems

If $\boldsymbol{x}^*$ is a local minimizer of (1), then there exists some $\boldsymbol{y} \geq \boldsymbol{0}$ with

$$\nabla f(\boldsymbol{x}^*) + \boldsymbol{A}^\top \boldsymbol{y} = \boldsymbol{0}$$
$$y_i \cdot (\boldsymbol{a}_i^\top \boldsymbol{x}^* - b_i) = 0 \quad \forall \; i,$$

where $\boldsymbol{a}_i^\top$ is the ith row of $\boldsymbol{A}$.

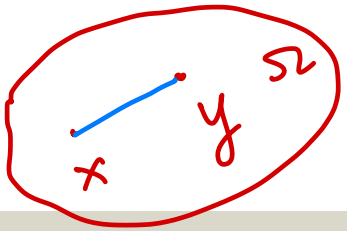
Remark:

- This can be extended to include equality constraints.
- This is a special case of KKT condition with linear constraints. It does not need constraint qualifications to be necessary.

Review

- Optimality conditions for nonlinear optimization.
- Convexity.
  - Convex set.
  - Convex function.
  - Convex optimization and optimality conditions.
- Algorithms.
- Integer optimization

Convex Sets



Definition: Convex Set

Ω is a convex set if and only if for any $\boldsymbol{x}, \boldsymbol{y} \in \Omega$ and any $0 \leq \lambda \leq 1$: $\lambda \boldsymbol{x} + (1 - \lambda)\boldsymbol{y} \in \Omega$.

Remark:

- To check whether a set is a convex set or not, most of the time, just check the definition.

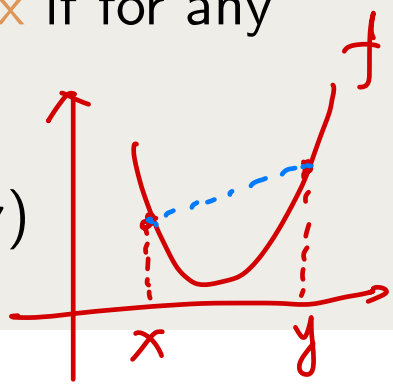
- Optimality conditions for nonlinear optimization.
- Convexity.
  - Convex set.
  - Convex function.
  - Convex optimization and optimality conditions.
- Algorithms.
- Integer optimization

Convex Functions

Definition: Convex Function

A function f on a convex set Ω is said to be convex if for any $\boldsymbol{x}, \boldsymbol{y} \in \Omega$ and any $0 \leq \lambda \leq 1$,

$$f(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}) \leq \lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y})$$



Definition: Concave Function

We call f a concave function if and only if $-f$ is a convex function or for any $\boldsymbol{x}, \boldsymbol{y} \in \Omega$ and $0 \leq \lambda \leq 1$,

$$f(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}) \geq \lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y}).$$

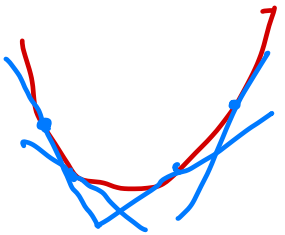
- You should be able to check whether a function is convex or concave or neither.

Equivalent Definitions

Theorem: Convexity via Hessian

Let f be twice differentiable over a convex open set Ω. Then f is convex on Ω if and only if its Hessian matrix is positive semidefinite

At any point $x \in \Omega$



1-D: $f'(x)$ is non-decreasing.

Equivalent Definitions

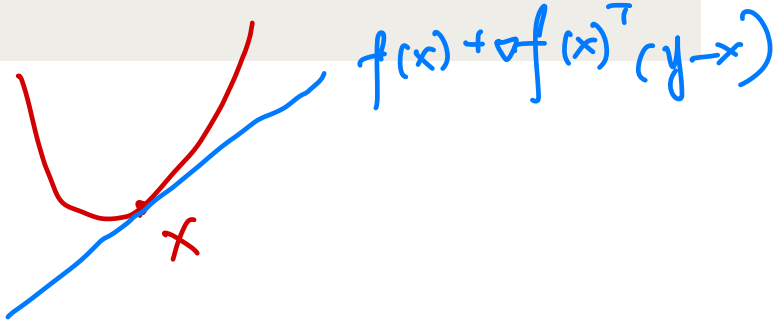
Theorem: Convexity via Hessian

Let f be twice differentiable over a convex open set Ω. Then f is convex on Ω if and only if its Hessian matrix is positive semidefinite

Theorem: First-Order Characterization of Convexity

Let f be differentiable over a convex open set Ω. Then f is convex if and only if

$$f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^\top (\boldsymbol{y} - \boldsymbol{x}), \quad \forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$$



Lemma: Sum Rule

If $a_1, \ldots, a_m \geq 0$, and $f_1, \ldots, f_m$ are convex functions, then $a_1 f_1 + \cdots + a_m f_m$ is a convex function.

Lemma: Sum Rule

If $a_1, \ldots, a_m \geq 0$, and $f_1, \ldots, f_m$ are convex functions, then $a_1 f_1 + \cdots + a_m f_m$ is a convex function.

Lemma: Composition with Linear Functions

If $f : \mathbb{R}^m \to \mathbb{R}$ is convex and $\boldsymbol{A} \in \mathbb{R}^{m \times n}$, $\boldsymbol{b} \in \mathbb{R}^m$ are given, then $g : \mathbb{R}^n \to \mathbb{R}$, $g(\boldsymbol{x}) := f(\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b})$, is convex.

Lemma: Sum Rule

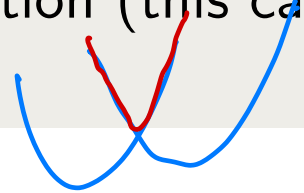
If $a_1, \ldots, a_m \geq 0$, and $f_1, \ldots, f_m$ are convex functions, then $a_1 f_1 + \cdots + a_m f_m$ is a convex function.

Lemma: Composition with Linear Functions

If $f : \mathbb{R}^m \to \mathbb{R}$ is convex and $\boldsymbol{A} \in \mathbb{R}^{m \times n}$, $\boldsymbol{b} \in \mathbb{R}^m$ are given, then $g : \mathbb{R}^n \to \mathbb{R}$, $g(\boldsymbol{x}) := f(\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b})$, is convex.

Lemma: Taking Maximum

If $f_1, \ldots, f_m$ are convex functions, then $f(\boldsymbol{x}) = \max\{f_1(\boldsymbol{x}), \ldots, f_m(\boldsymbol{x})\}$ is a convex function (this can be extended to uncountably many).



Lemma: Composition

Let $\Omega \subset \mathbb{R}^n$ be convex. Suppose $g : \Omega \to \mathbb{R}$ is convex, and $f : I \to \mathbb{R}$ is convex and non-decreasing, where $I \supseteq g(\Omega)$ is an interval containing $g(\Omega)$. Then, $f \circ g$ is convex.

Lemma: Composition

Let $\Omega \subset \mathbb{R}^n$ be convex. Suppose $g : \Omega \to \mathbb{R}$ is convex, and $f : I \to \mathbb{R}$ is convex and non-decreasing, where $I \supseteq g(\Omega)$ is an interval containing $g(\Omega)$. Then, $f \circ g$ is convex.

Proof:

For all $x, y \in \Omega$ and $\theta \in [0, 1]$, the convexity of g implies

$$g(\theta x + (1-\theta)y) \leq \theta g(x) + (1-\theta)g(y).$$

Since f is non-decreasing, means $a \leq b \Rightarrow f(a) \leq f(b)$

$$f(g(\theta x + (1-\theta)y)) \leq f(\theta g(x) + (1-\theta)g(y)).$$

Finally, the convexity of f ensures

$$f(\theta g(x) + (1-\theta)g(y)) \leq \theta f(g(x)) + (1-\theta)f(g(y)).$$

Combining these results, we obtain

$$f(g(\theta x + (1-\theta)y)) \leq \theta f(g(x)) + (1-\theta)f(g(y)).$$

Remark: if f, g are convex, $f \circ g$ may not be convex.

Example $f(x) = -x$

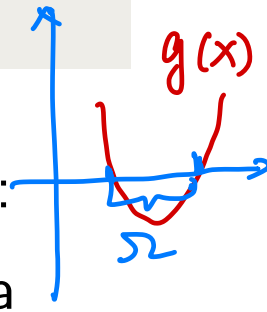
$g(x) = x^2$

$f \circ g(x) = -x^2$ not convex.

- Optimality conditions for nonlinear optimization.
- Convexity.
  - Convex set.
  - Convex function.
  - Convex optimization and optimality conditions.
- Algorithms.
- Integer optimization

Definition: Convex Constraint

A constraint is called convex if the feasible region defined by the constraint is a convex set.



In practice, we frequently use the following sufficient condition:

- Constraints of the form $g(\boldsymbol{x}) \leq 0$ with g being convex is a convex constraint.
- Linear constraints $\boldsymbol{Ax} \leq \boldsymbol{b}$, $\boldsymbol{Ax} = \boldsymbol{b}$ and $\boldsymbol{Ax} \geq \boldsymbol{b}$, are convex constraints.
- Combinations of above constraints are convex constraints.

Not nesessary condition, e.g.,

$\{x : -x^2 \leq 1\} = \mathbb{R}$ still convex

Convex Optimization Problems

A convex optimization problem is a problem:

- Minimize a convex function or maximize a concave function.
- The constraints are all convex constraints.

Convex Optimization Problems

A convex optimization problem is a problem:

- Minimize a convex function or maximize a concave function.
- The constraints are all convex constraints.

Sometimes, we need to apply some transformations to make a problem convex.

Common Transformations:

- Monotone transformation of the objective function and/or the constraints. e.g. $\log(\cdot)$
- Substitute variables to remove nonlinearity.

You should be able to determine whether an optimization problem is convex, and find ways to reformulate a nonconvex problem as a convex problem if it's possible.

Convexity, Stationarity and Global Optimality

Theorem: Convexity and Global Solutions

Let $f : \Omega \to \mathbb{R}$ be a convex function and $\Omega \subseteq \mathbb{R}^n$ be a convex set. Then any local minimizer of the problem: minimize$_{\boldsymbol{x} \in \Omega} f(\boldsymbol{x})$ is a global minimizer.

Convexity, Stationarity and Global Optimality

Theorem: Convexity and Global Solutions

Let $f : \Omega \to \mathbb{R}$ be a convex function and $\Omega \subseteq \mathbb{R}^n$ be a convex set. Then any local minimizer of the problem: $\text{minimize}_{\boldsymbol{x} \in \Omega} f(\boldsymbol{x})$ is a global minimizer.

Theorem: Stationarity & Global Optimality

Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is convex on $\mathbb{R}^n$ and continuously differentiable at $\boldsymbol{x}^*$. Then, $\boldsymbol{x}^*$ is a global minimizer of $\min_{\boldsymbol{x} \in \mathbb{R}^n} f(\boldsymbol{x})$ if and only if $\nabla f(\boldsymbol{x}^*) = 0$.

Convexity, Stationarity and Global Optimality

Theorem: Convexity and Global Solutions

Let $f : \Omega \to \mathbb{R}$ be a convex function and $\Omega \subseteq \mathbb{R}^n$ be a convex set. Then any local minimizer of the problem: $\text{minimize}_{\boldsymbol{x} \in \Omega} f(\boldsymbol{x})$ is a global minimizer.

Theorem: Stationarity & Global Optimality (unconstrained)

Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is convex on $\mathbb{R}^n$ and continuously differentiable at $\boldsymbol{x}^*$. Then, $\boldsymbol{x}^*$ is a global minimizer of $\min_{\boldsymbol{x} \in \mathbb{R}^n} f(\boldsymbol{x})$ if and only if $\nabla f(\boldsymbol{x}^*) = 0$.

Remarks:

- In unconstrained convex optimization: stationarity (FONC) is sufficient and necessary for global optimality.
- Finding stationary points of a convex optimization problems providing global optimal solutions.

KKT Conditions is Sufficient

Theorem: Stationarity & Global Optimality

Consider the constrained convex optimization problem

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \le 0, \quad \forall\ i = 1,\dots,m, \\ & h_j(\boldsymbol{x}) = 0, \quad \forall\ j = 1,\dots,p, \end{aligned}$$

where f and $\{g_i\}_{i=1}^m$ are convex, and $\{h_j\}_{j=1}^p$ are linear. Then any KKT point for this problem is a global minimizer.

$\Omega = \{x : g_i(x) \le 0,\ h_j(x) = 0\}$ is convex

because g_i is convex and h_j is linear.

KKT Conditions is Sufficient

Theorem: Stationarity & Global Optimality

Consider the constrained convex optimization problem

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \le 0, \quad \forall\ i = 1, \dots, m, \\ & h_j(\boldsymbol{x}) = 0, \quad \forall\ j = 1, \dots, p, \end{aligned}$$

where f and $\{g_i\}_{i=1}^m$ are convex, and $\{h_j\}_{j=1}^p$ are linear. Then any KKT point for this problem is a global minimizer.

Remarks:

- If f ~~and Ω are convex~~ and g_i convex and h_j linear, then KKT conditions (FONC) are sufficient for global minimizers (no need of CQ at all).
- Recall that in order to let the necessary direction (KKT conditions) to be true, we need CQ to hold.

Review

- Optimality conditions for nonlinear optimization.
- Convexity.
- Algorithms.
  - Bisection method and golden section method.
  - Gradient descent method.
  - Newton's method.
  - Projected gradient descent method.

1-D

higher dimension

unconstrained

constrained problem.

- Integer optimization

Bisection Method:

- Try to find the root of f': find x with $f'(x) = 0$.
- We need to know the derivative.

Golden Section Method: → for function that has only one local min/local max inside the interval.

- The method is directly applied to find the optimal solution.
- We do not need derivative information.

Review

- Optimality conditions for nonlinear optimization.
- Convexity.
- Algorithms.
  - Bisection method and golden section method.
  - Gradient descent method.
  - Newton's method.
  - Projected gradient descent method.
- Integer optimization

An Algorithm Framework:

Start from a point $\boldsymbol{x}^0$, set a tolerance level $\varepsilon > 0$ and $k = 0$.

1. If $||\nabla f(\boldsymbol{x}^k)|| < \varepsilon$, then output $\boldsymbol{x}^k$, otherwise continue with step 2.
2. Compute the search direction $\boldsymbol{d}^k$. (usually d^k is a descent direction)
3. Choose step size α_k by backtracking or exact line search.
4. Set $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k$.

In the gradient descent method:

$$\boldsymbol{d}^k = -\nabla f(\boldsymbol{x}^k).$$

← is dsent direction of x^k is not sationary. because $\nabla f(x^k)^T d^k = -\|\nabla f(x^k)\|^2$

That is

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \alpha_k \nabla f(\boldsymbol{x}^k).$$

Exact Line Search:

Choose α_k to achieve the largest descent:

$$\alpha_k = \text{argmin}_{\alpha \geq 0} f(\boldsymbol{x}^k + \alpha \boldsymbol{d}^k).$$

If f is quadratic, we have explicit formula for α_k.

Exact Line Search:

Choose α_k to achieve the largest descent:

$$\alpha_k = \text{argmin}_{\alpha \geq 0} f(\boldsymbol{x}^k + \alpha \boldsymbol{d}^k).$$

Backtracking / Armijo Line Search:

Let $\sigma, \gamma \in (0, 1)$ be given. Choose α_k as the largest element in $\{1, \sigma, \sigma^2, \sigma^3, \dots\}$ such that

$$f(\boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k) - f(\boldsymbol{x}^k) \leq \gamma \alpha_k \cdot \nabla f(\boldsymbol{x}^k)^\top \boldsymbol{d}^k.$$

- This condition is called Armijo condition.
- α_k can be determined after finitely many steps if $\boldsymbol{d}^k$ is a descent direction.

Convergence of the Gradient Method

Global Convergence:

- With proper step sizes, the sequence $\{f(\boldsymbol{x}^k)\}_k$ is nonincreasing and every accumulation point of $\{\boldsymbol{x}^k\}_k$ is a stationary point of f.

Linear Convergence under Additional Assumptions:

- Let f be strongly convex with constant μ and Lipschitz smooth with constant L. With proper step sizes, $\{\boldsymbol{x}_k\}_k$ converges linearly to the global solution $\boldsymbol{x}^*$ or $\{f(\boldsymbol{x}_k)\}_k$ converges linearly to $f(\boldsymbol{x}^*)$.

Review the definition for linear convergence.

Review

- Optimality conditions for nonlinear optimization.
- Convexity.
- Algorithms.
  - Bisection method and golden section method.
  - Gradient descent method.
  - Newton's method.
  - Projected gradient descent method.
- Integer optimization

Idea: The second-order Taylor expansion:

$$f(\boldsymbol{x}) \approx f(\boldsymbol{x}^k) + \nabla f(\boldsymbol{x}^k)^\top (\boldsymbol{x} - \boldsymbol{x}^k) + \frac{1}{2}(\boldsymbol{x} - \boldsymbol{x}^k)^\top \nabla^2 f(\boldsymbol{x}^k)(\boldsymbol{x} - \boldsymbol{x}^k)$$

Newton's Method

Idea: The second-order Taylor expansion:

$$f(\boldsymbol{x}) \approx f(\boldsymbol{x}^k) + \nabla f(\boldsymbol{x}^k)^\top (\boldsymbol{x} - \boldsymbol{x}^k) + \frac{1}{2}(\boldsymbol{x} - \boldsymbol{x}^k)^\top \nabla^2 f(\boldsymbol{x}^k)(\boldsymbol{x} - \boldsymbol{x}^k)$$

Newton's Method Iteration:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - (\nabla^2 f(\boldsymbol{x}^k))^{-1} \nabla f(\boldsymbol{x}^k).$$

This corresponds to the minimizer of the quadratic approximation, provided $\nabla^2 f(\boldsymbol{x}^k)$ is positive definite.

Newton's Method

Idea: The second-order Taylor expansion:

$$f(\boldsymbol{x}) \approx f(\boldsymbol{x}^k) + \nabla f(\boldsymbol{x}^k)^\top (\boldsymbol{x} - \boldsymbol{x}^k) + \frac{1}{2}(\boldsymbol{x} - \boldsymbol{x}^k)^\top \nabla^2 f(\boldsymbol{x}^k)(\boldsymbol{x} - \boldsymbol{x}^k)$$

Newton's Method Iteration:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - (\nabla^2 f(\boldsymbol{x}^k))^{-1} \nabla f(\boldsymbol{x}^k).$$

This corresponds to the minimizer of the quadratic approximation, provided $\nabla^2 f(\boldsymbol{x}^k)$ is positive definite.

Idea: The second-order Taylor expansion:

$$f(\boldsymbol{x}) \approx f(\boldsymbol{x}^k) + \nabla f(\boldsymbol{x}^k)^\top (\boldsymbol{x} - \boldsymbol{x}^k) + \frac{1}{2}(\boldsymbol{x} - \boldsymbol{x}^k)^\top \nabla^2 f(\boldsymbol{x}^k)(\boldsymbol{x} - \boldsymbol{x}^k)$$

Newton's Method Iteration:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - (\nabla^2 f(\boldsymbol{x}^k))^{-1} \nabla f(\boldsymbol{x}^k).$$

This corresponds to the minimizer of the quadratic approximation, provided $\nabla^2 f(\boldsymbol{x}^k)$ is positive definite.

Search Direction (Newton's Direction):

$$\boldsymbol{d}^k = -(\nabla^2 f(\boldsymbol{x}^k))^{-1} \nabla f(\boldsymbol{x}^k).$$

Convergence of the Pure Newton's Method:

- "Hessian uniformly positive definite + Lipschitz continuous" and "$\boldsymbol{x}^0$ close to $\boldsymbol{x}^*$" local convergence

  $$\Longrightarrow \quad \{\boldsymbol{x}^k\}_k \text{ converges quadratically to } \boldsymbol{x}^*,$$

  i.e., $\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^*\| \leq C\|\boldsymbol{x}^k - \boldsymbol{x}^*\|^2$.

Convergence of the Pure Newton's Method:

- "Hessian uniformly positive definite + Lipschitz continuous" and "$\boldsymbol{x}^0$ close to $\boldsymbol{x}^*$"

$$\Longrightarrow \quad \{\boldsymbol{x}^k\}_k \text{ converges quadratically to } \boldsymbol{x}^*,$$

i.e., $\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^*\| \leq C\|\boldsymbol{x}^k - \boldsymbol{x}^*\|^2$.

Backtracking with Newton's Direction:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \alpha_k(\nabla^2 f(\boldsymbol{x}^k))^{-1}\nabla f(\boldsymbol{x}^k).$$

- Choose α_k by backtracking line search.
- This version converges globally: every acc. point is a stationary point — independent of the initial point.

Tradeoff between Newton and Gradient Methods

- Gradient method is simple and fast in each iteration and needs less storage. However, it takes much more iterations to converge.
- Newton's method takes much less iterations to converge. However, each iteration is more expensive in both computation and storage space.

Review

- Optimality conditions for nonlinear optimization.
- Convexity.
- Algorithms.
  - Bisection method and golden section method.
  - Gradient descent method.
  - Newton's method.
  - Projected gradient descent method.
- Integer optimization

Projected Gradient Method

$\min_{x \in \Omega} f(x)$

Idea: The step $\boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k$ might not be feasible, so we project the step back onto the feasible set.

Definition: Euclidean Projections

Let $\Omega \subset \mathbb{R}^n$ be a nonempty, closed, convex set. The (Euclidean) pro- jection $\mathcal{P}_\Omega(\boldsymbol{x})$ of $\boldsymbol{x}$ onto Ω is unique minimizer of:

$$\min_{\boldsymbol{y}} \ \frac{1}{2}\|\boldsymbol{x} - \boldsymbol{y}\|^2 \quad \text{s.t.} \quad \boldsymbol{y} \in \Omega.$$

Iteration:

$$\boldsymbol{x}^{k+1} = \mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)).$$

Backtracking Line Search for Projected Gradient

If λ_k is determined by line search, every adjustment of λ_k requires to reevaluate $\mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k))$. This can be expensive.

Backtracking Line Search for Projected Gradient

If λ_k is determined by line search, every adjustment of λ_k requires to reevaluate $\mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k))$. This can be expensive.

Observation:

$$\boldsymbol{x}^{k+1} = \mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) = \boldsymbol{x}^k + [\mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) - \boldsymbol{x}^k].$$

Backtracking Line Search for Projected Gradient

If λ_k is determined by line search, every adjustment of λ_k requires to reevaluate $\mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k))$. This can be expensive.

Observation:

$$\boldsymbol{x}^{k+1} = \mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) = \boldsymbol{x}^k + [\mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) - \boldsymbol{x}^k].$$

Consider $\boldsymbol{d}^k = \mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) - \boldsymbol{x}^k$ as a search direction.

Backtracking Line Search for Projected Gradient

If λ_k is determined by line search, every adjustment of λ_k requires to reevaluate $\mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k))$. This can be expensive.

Observation:

$$\boldsymbol{x}^{k+1} = \mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) = \boldsymbol{x}^k + [\mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) - \boldsymbol{x}^k].$$

Consider $\boldsymbol{d}^k = \mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) - \boldsymbol{x}^k$ as a search direction. Introduce another step size α_k:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k = (1 - \alpha_k)\boldsymbol{x}^k + \alpha_k \mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)).$$

Backtracking Line Search for Projected Gradient

If λ_k is determined by line search, every adjustment of λ_k requires to reevaluate $\mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k))$. This can be expensive.

Observation:

$$\boldsymbol{x}^{k+1} = \mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) = \boldsymbol{x}^k + [\mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) - \boldsymbol{x}^k].$$

Consider $\boldsymbol{d}^k = \mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) - \boldsymbol{x}^k$ as a search direction. Introduce another step size α_k:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k = (1 - \alpha_k)\boldsymbol{x}^k + \alpha_k \mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)).$$

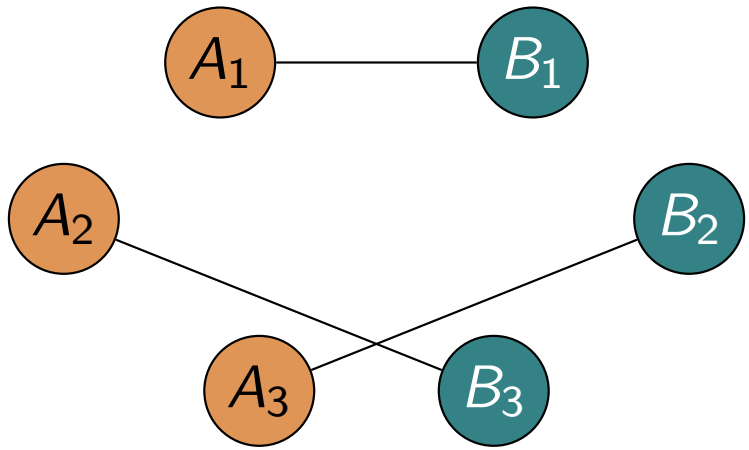
- If $\alpha_k \in [0, 1]$, then the convexity of Ω implies that $\boldsymbol{x}^{k+1}$ will be feasible if $\boldsymbol{x}^k \in \Omega$.
- We can do backtracking line search on α_k.

- Optimality conditions for nonlinear optimization.
- Convexity.
- Algorithms.
- Integer optimization
  - Modeling.
  - LP relaxation.
  - Branch and bound.

Example: Matching Problem

Setup: Assume there are two groups of individuals registered on a dating platform, labeled Group A and Group B. Each group consists of n individuals.

- Each individual i from Group A rates each individual j from Group B with a score v_{ij}.
- The goal is to find a one-to-one matching between the groups that maximizes the total score.



Matching Problem

Decision Variables:

- x_{ij}: Binary variable indicating whether individual i from Group A is matched with individual j from Group B.

Objective Function:

$$\text{maximize} \quad \sum_{i=1}^{n} \sum_{j=1}^{n} v_{ij} x_{ij}$$

Matching Problem

Decision Variables:

- x_{ij}: Binary variable indicating whether individual i from Group A is matched with individual j from Group B.

Objective Function:

$$\text{maximize} \quad \sum_{i=1}^{n}\sum_{j=1}^{n} v_{ij}x_{ij}$$

Constraints:

- Each individual in Group A is matched to exactly one person in Group B: $\sum_{j=1}^{n} x_{ij} = 1, \quad \forall i \in \{1, \ldots, n\}$.
- Each individual in Group B is matched to exactly one person in Group A: $\sum_{i=1}^{n} x_{ij} = 1, \quad \forall j \in \{1, \ldots, n\}$.
- Binary assignment constraint: $x_{ij} \in \{0, 1\}, \quad \forall i, j$.

Matching Problem: Full Formulation

$$\begin{aligned} \text{maximize} \quad & \sum_{i=1}^{n}\sum_{j=1}^{n} v_{ij}x_{ij} \\ \text{s.t.} \quad & \sum_{i=1}^{n} x_{ij} = 1 \quad \forall\, j, \quad \sum_{j=1}^{n} x_{ij} = 1 \quad \forall\, i \\ & x_{ij} \in \{0,1\} \end{aligned}$$

It is a binary optimization problem.

Integer Optimization

Overview:

- Formulating integer problems.
- Role of LP relaxations in IP.
- Branch-and-bound algorithm.

- Optimality conditions for nonlinear optimization.
- Convexity.
- Algorithms.
- Integer optimization
  - Modeling.
  - LP relaxation.
  - Branch and bound.

LP Relaxation

Linear programming (LP) relaxation:

- Relax (ignore) all integer constraints and solve the associated linear program.
- Relaxation of binary constraints: $x \in \{0, 1\} \rightsquigarrow 0 \leq x \leq 1$.

Linear programming (LP) relaxation:

- Relax (ignore) all integer constraints and solve the associated linear program.
- Relaxation of binary constraints: $x \in \{0, 1\} \rightsquigarrow 0 \leq x \leq 1$.
- Provide bounds on the optimal value of IP (upper bound for maximization and lower bound for minimization).
- Integrality gap:

$$|v^{LP} - v^{IP}|$$

↓

because LP relaxation considers a larger feasible region.

LP Relaxation

Usually solving the LP relaxation does not give the optimal solution to the IP:

- Simply rounding a fractional solution to an integer solution does often not work.
- However, if the optimal solution of the LP relaxation is an integer point, then it is an optimal solution to IP as well.

LP Relaxation

Usually solving the LP relaxation does not give the optimal solution to the IP:

- Simply rounding a fractional solution to an integer solution does often not work.
- However, if the optimal solution of the LP relaxation is an integer point, then it is an optimal solution to IP as well.
- A sufficient condition to guarantee that an LP always has an integer optimal solution — Total Unimodularity (TU).

If the constraint matrix $\boldsymbol{A}$ is TU and $\boldsymbol{b}$ is an integer vector, then:

- All basic feasible solutions are integers.
- Solving the LP via the simplex method will solve the IP.

- Optimality conditions for nonlinear optimization.
- Convexity.
- Algorithms.
- Integer optimization
  - Modeling.
  - LP relaxation.
  - Branch and bound.

Branch-and-Bound Method

High-level Idea:

- **Branching:** Divide the feasible region into smaller ones, solve each of them and combine them to find the optimal solution.
- **Bounding:** Use bounds (LP optimal value and feasible integer points) to reduce the number of branches we need to consider.

Branching Procedures:

1. Solve the LP relaxation.
   - If the optimal solution is integral, then it is optimal to IP.
   - Otherwise go to step 2.
2. If the optimal solution to the LP relaxation is $\boldsymbol{x}^*$ and x_i^* is fractional, then branch the problem into the following two:
   - One with an added constraint that $x_i \leq \lfloor x_i^* \rfloor$.
   - One with an added constraint that $x_i \geq \lceil x_i^* \rceil$.
3. For each of the two problems, use the same method to solve them, and get optimal sol. $\boldsymbol{y}_1^*$ and $\boldsymbol{y}_2^*$ with optimal value v_1^* and v_2^*.
   - Compare to obtain the optimal solution.

Bounding procedures (for maximization):

- Any LP relaxation solution can provide an upper bound for each node in the branching process.
- Any feasible point to the IP can provide a lower bound for the entire problem.

upper bound for current node/subproblem

every time we get an integer solution for a LP relaxation of a subproblem, it's also a feasible solution for original IP. Therefore, it provides a lower bound for original IP.

When the optimal value of the LP relaxation of this branch is less than (or no better than) the current lower bound, then we can discard this branch.

- No better solution can be obtained from further exploring this branch.

Course and Teaching Evaluation



CTE